

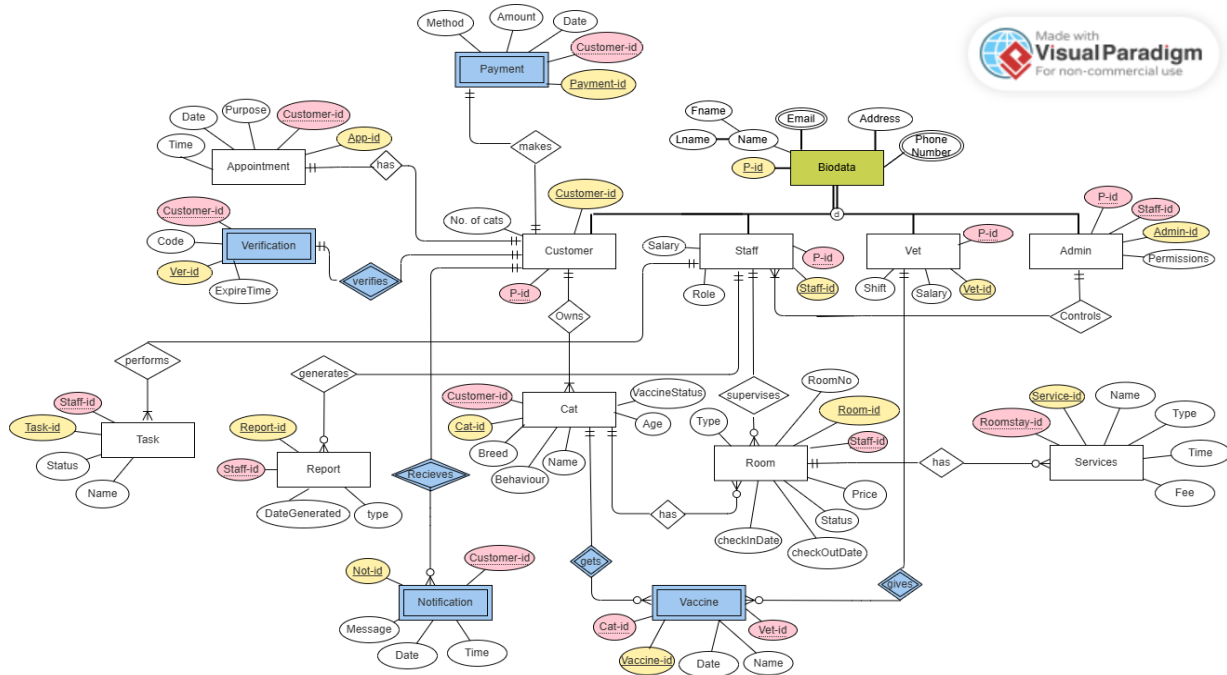
DATABASE MANAGEMENT SYSTEM
RDM + NORMALIZATION
CAT RESORT MANAGEMENT SYSTEM



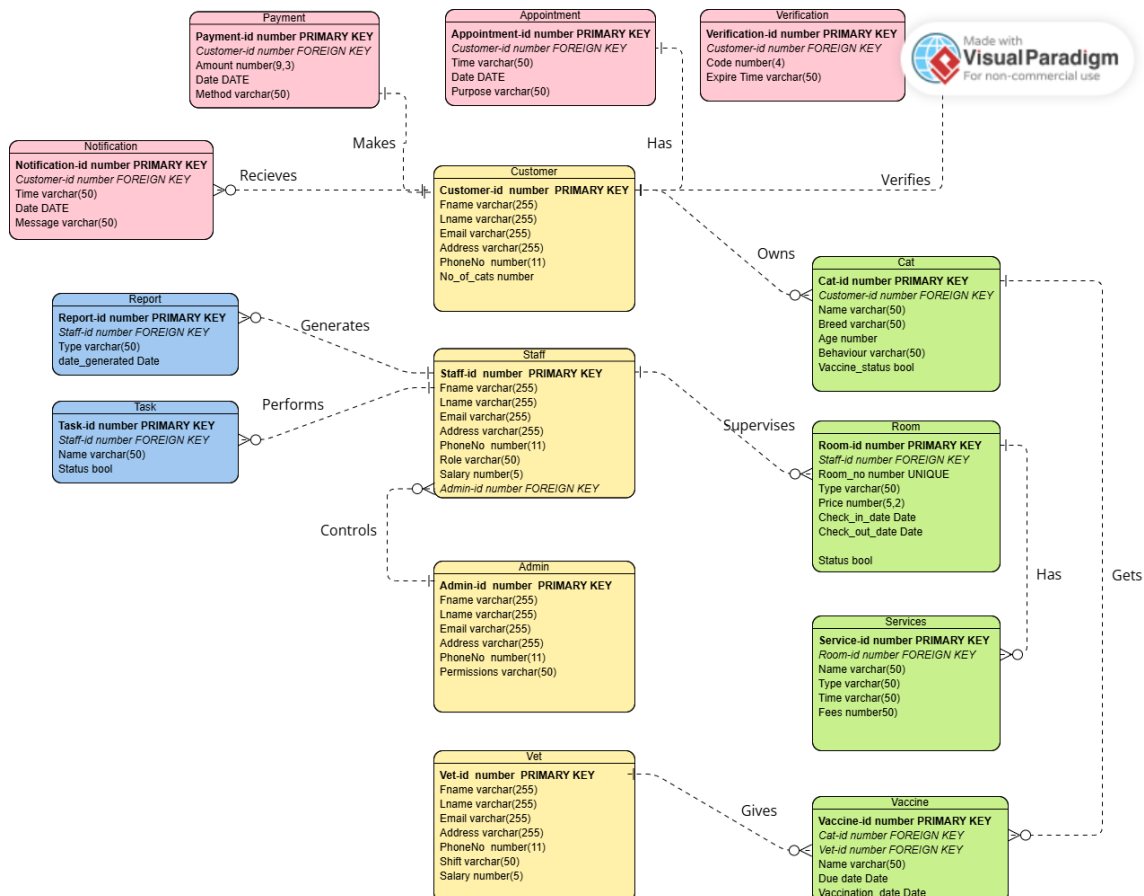
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Entity-Relationship Diagram



Relational Data Model



NORMALIZATION

1. Customer

Customer	<u>Customer-id</u>	Fname	Lname	Address	Email	Phone Number	No of Cats
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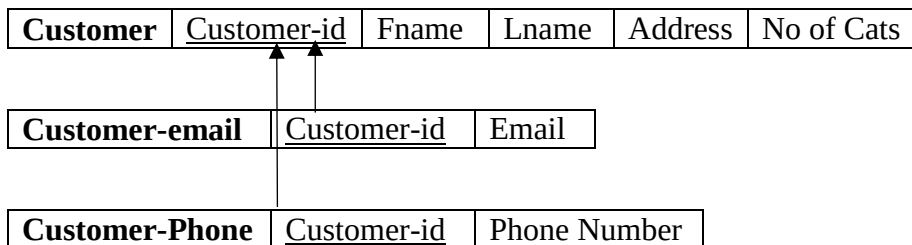
1NF:

- Each attribute must contain atomic values.
- Every record must be unique.

Problem in our table:

- Email and phone_number are multivalued.

Solution: Split email and phone_number into separate tables or make each row atomic.



2NF:

- Must be in 1NF.
- Every non-key attribute must be fully functionally dependent on the primary key.
- Remove partial dependencies.

Check:

- Customer table primary key: customer_id
- Non-key attributes: fname, lname, address, no_of_cats
- No partial dependency here since customer_id is a single attribute.

Customer_Email and Customer_Phone are already dependent on customer_id.

3NF:

- Must be in 2NF.
- **No transitive dependencies:** non-key attributes cannot depend on other non-key attributes.

Check Customer Table:

- customer_id → fname, lname, address, no_of_cats
- Any attribute depending on another non-key attribute?
 - no_of_cats could be derived from a separate table Cat, which would store individual cats.

Solution: Split cats into a separate table.

2. Admin

Admin	<u>Admin-id</u>	Fname	Lname	Address	Phone Number	Email	Permissions
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1NF: Split multivalued attributes: email and phone_number

Admin	<u>Admin-id</u>	Fname	Lname	Address	Permissions
Admin-email	<u>Admin-id</u>	Email			
Admin-Phone	<u>Admin-id</u>	Phone Number			

2NF:

- Primary key: admin_id
- Non-key attributes fully dependent on admin_id, already 2NF

3NF: No transitive dependencies, 3NF achieved

3. Staff

Staff	<u>Staff-id</u>	Fname	Lname	Email	Address	Phone Number	Role	Salary	Admin-id
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1NF: Split multivalued attributes: email and phone_number

Staff	<u>Staff-id</u>	Fname	Lname	Address	Role	Salary	Admin-id
Staff-email	<u>Staff-id</u>	Email					
Staff-phone	<u>Staff-id</u>	Phone Number					

2NF:

- Primary key: staff_id (single attribute)
- No partial dependencies, already in 2NF

3NF: admin_id is a foreign key; any admin info stored in this table would create transitive dependency, move admin details to Admin table, **3NF achieved**

4. Vet

Vet	<u>Vet-id</u>	Fname	Lname	Email	Address	Phone Number	Shift	Salary
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1NF: Split multivalued attributes: email and phone_number

Vet	<u>Vet-id</u>	Fname	Lname	Address	Shift	Salary
Vet-email	<u>Vet-id</u>	Email				
Vet-Phone	<u>Vet-id</u>	Phone Number				

2NF:

- Primary key: vet_id
- Non-key attributes fully dependent on vet_id, already 2NF

3NF: No transitive dependencies, 3NF achieved

5. Task

Task	<u>Task-id</u>	Staff-id	Name	Status
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1NF: Each attribute must contain atomic values and every record must be unique.

No changes needed.

2NF: Already in 2NF

- Primary Key: Task-id (single attribute)
- Non-key attributes: name, status, staff-id
- All depend fully on Task-id
- No partial dependency

3NF: No transitive dependencies, 3NF achieved

6. Cat

Cat	<u>Cat-id</u>	Customer-id	Name	Breed	Age	Behavior	Vaccine-Status
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1NF: Values are atomic and no multivalued attribute.

2NF: Already in 2NF

- Primary Key: Cat-id
- All attributes depend fully on Cat-id.
- No partial dependency.

3NF

- Vaccine status depends on vaccine table, not directly Cat-id
- It is a derived attribute (cat gets multiple vaccines)

Problem: Vaccine status should not be stored in Cat table.

Solution: Remove Vaccine status.

Cat	<u>Cat-id</u>	Customer-id	Name	Breed	Age	Behavior	Vaccine-Status
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Cat	Cat-id	Customer-id	Name	Breed	Age	Behaviour
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7. Room

Room	<u>Room-id</u>	Staff-id	Room-no	Type	Price	Check-in-date	Check-out-date	Status
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1NF: Values are atomic and no multivalued attribute.

2NF:

Primary Key: Room-id, all non-key attributes depend on Room-id, no partial dependency.

3NF:

- staff_id is a foreign key
- room_no is unique but not functionally determining other fields in RDM.
- status depends only on room usage, not other non-key attributes

No transitive dependency.

8. Services

Services	<u>Service-id</u>	Room-id	Name	Type	Time	Fees
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1NF: Values are atomic and no multivalued attribute.

2NF: Primary Key: Service-id. All attributes depend fully on Service-id. No partial dependency. 2NF achieved.

3NF: room_id does not determine name, type, time, or fees, fees do not depend on any other non-key. No transitive dependency.

9. Vaccine

Vaccine	<u>Vaccine-id</u>	Cat-id	Vet-id	Name	Due-date	Vaccination-date	Vaccine-Status
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1NF: All values atomic, no multivalued attributes.

2NF: Primary Key: Vaccine-id. All other attributes depend fully on Vaccine-id. 2NF satisfied.

3NF: cat_id and vet_id are foreign keys, name, due_date, vaccination_date depend only on vaccine_id. No transitive dependency.

10. Notification

Notification	<u>Notification_id</u>	Customer-id	message	date	time
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1NF: Each attribute must contain atomic values. Every record must be unique.

No change needed because there are no repeating values.

2NF: Must be in 1NF. Non-key attributes fully dependent on not-id, already 2NF

3NF: Must be in 2NF. No **transitive dependency** (non-key attribute $\rightarrow$ non-key attribute). There is no attribute that depends on Customer-id.

11.Payment

Payment	<u>Payment-id</u>	customer-id	date	Amount	Method
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1NF: Each attribute must contain atomic values. Every record must be unique. No change needed because there are no repeating values.

2NF: Must be in 1NF. Non-key attributes fully dependent on Payment-id, already 2NF

3NF: Must be in 2NF. No **transitive dependency** (non-key attribute $\rightarrow$ non-key attribute)

Check dependencies:

- Payment-id $\rightarrow$ Customer-id
- Payment-id $\rightarrow$ Method
- Payment-id $\rightarrow$ Amount
- Payment-id $\rightarrow$ Date

There is no attribute that depends on Method and there is no attribute that depends on Customer-id.

12.Appointment

Appointment	<u>Appointment-id</u>	Customer-id	time	date	purpose
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1NF: Each attribute must contain atomic values. Every record must be unique. No change needed because there are no repeating values.

2NF: Must be in 1NF. Non-key attributes fully dependent on appointment-id, already 2NF

3NF: Table must be in 2NF. No **transitive dependencies** (non-key attributes must NOT depend on other non-key attributes)

Check:

- Time $\rightarrow$ depends only on Appointment-id
- Date $\rightarrow$ depends only on Appointment-id
- Purpose $\rightarrow$ depends only on Appointment-id

No attribute depends on Customer-id or another non-key attribute.

13.Verification

Verification	<u>Verification-id</u>	Customer-id	Code-number	Expire-time
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1NF: No change needed because there are no repeating values.

2NF: Must be in 1NF. Non-key attributes fully dependent on appointment-id, already 2NF

3NF: Table must be in 2NF. No **transitive dependencies** (non-key attributes must NOT depend on other non-key attributes)

- Code depends only Verification-id
- Expire Time depends only on Verification-id
- Customer-id is only a foreign key

No attribute depends on another non-key attribute.

14.Report

Report	<u>Report-id</u>	Staff-id	type	Date_generated
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1NF: All values must be atomic. No repeating groups. Unique rows

The Report table is already in 1NF

2NF: Table is in 1NF. No partial dependencies (only applies if PK is composite)

Here, the primary key = Report-id (single attribute)

Therefore: No partial dependencies possible. Every attribute depends on the full key

3NF: Table is in 2NF. No transitive dependencies

Check dependencies:

- Staff-id → (does not determine any other non-key attribute in this table)
- Type → only depends on Report-id
- date_generated → only depends on Report-id